

SAMYAK JAIN

Manufacturing & Quality Engineer | NPI, APQP/PPAP, SPC/DMAIC, Analytics

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SUMMARY OF QUALIFICATIONS

- Manufacturing and quality engineer with automotive NPI launch experience (**APQP/PPAP** for Yamaha, Suzuki, and Piaggio) and data-driven process improvement that raised **Cpk from 0.46 to 1.49** and First-Pass Yield from 80% to 99.4%.
- Uses SQL and Power BI dashboards, FlexSim simulation, Gauge R&R, SPC, and DMAIC to find and fix production problems; **Six Sigma Green Belt**, ISO 9001 and IATF 16949 Internal Auditor; MS in Industrial Engineering and MBA.

EDUCATION

- **Texas A&M University, College Station, TX** (Master of Science in Industrial Engineering) **05/2026**
- **Narsee Monjee Institute of Management Studies, Maharashtra, India** (MBA) **02/2024**
- **Guru Gobind Singh Indraprastha University, Delhi, India** (BS in Mechanical and Automation Engineering) **05/2021**

WORK EXPERIENCE

Texas A&M University, Chemistry Department, College Station, TX: Lab Technician **01/2025 – 05/2026**

- Supported installation and maintenance of chemical fume hoods, coordinating with vendors and verifying airflow and **safety system performance** after commissioning.
- Led a **materials safety upgrade** replacing mercury bubblers with mercury-free alternatives across lab setups, evaluating technical requirements, sourcing compliant replacements, and completing the transition with zero disruption to ongoing operations.

Utility Power Systems, Delhi, India: Project Engineer **01/2023 – 07/2024**

- Built and presented status reviews for customer orders, scheduled with the critical path method (CPM); escalated at-risk steps and ran work in parallel to **pull in launch dates**, negotiated dates with customers, and **helped win additional orders**.
- Built the plant's **first real-time production dataset** in Odoo, with SQL and **Power BI dashboards** for OEE, First-Pass Yield, downtime, and on-time delivery refreshed every 5 to 10 minutes.
- Ran a **Gauge R&R** that traced 35-40% of variation to operator measurement, then used SPC and DMAIC to find the four process variables behind wall-thickness and ovality defects, raising **Cpk from 0.46 to 1.49** and eliminating \$150K/year in scrap.
- Used 80-85% utilization data to build a **FlexSim discrete-event simulation** capacity case that supported approval of a second pipe-bending machine, then issued **RFQs**, worked with the OEM on installation and validation, and trained operators.
- Used **Theory of Constraints** to target the pipe-bending bottleneck, applying SMED and planned maintenance to raise its output about 20% (bends per day), and wrote **SOPs and work instructions** for welding, blasting, galvanization, and powder coating.

Uno Minda Limited, Haryana, India: Manufacturing Engineer **01/2022 – 01/2023**

- Supported **APQP and PPAP** launches for Yamaha, Suzuki, and Piaggio, owning the control plan, sample parts, master sample, and checking aids, and presented **NPI, PPAP, and line-readiness** status for those parts in OEM launch reviews.
- Owned the process flow chart, floor plan layout, SOPs, error-proofing, and operator training for a new **pilot production line** that ran about **30% faster than the required line speed**, and ran the first pilot builds before launch.
- Flagged a **terminal mix-up risk** in PFMEA review and raised the engineering change that made the two terminal types impossible to swap, tested the new parts, and updated the control plan; also built **poka-yoke** into the wire-harness design.
- As **PFMEA action owner** on a PFMEA led by the engineering manager across 8 product designs, closed out recommended actions that caught 20+ potential failure modes and **cut design-stage issues about 30%** before tooling release.
- Managed **engineering changes in SAP** for 25+ product lines, keeping BOMs, routings, and production documents aligned with design intent, and self-initiated several ECRNs to make assembly easier for operators.

PROJECTS

- **Neural Networks for Manufacturing Inspection & Yield (2026)**: Built a CNN for automated visual defect inspection (ResNet18 transfer learning, 98.6% ROC-AUC, validated against ground-truth defect masks) and independently evaluated a semiconductor yield-prediction model, recommending against deployment after rigorous validation showed it performed no better than chance.
- **Advanced Quality Control Project (2025)**: Multivariate SPC (PCA, control charts) on high-dimensional manufacturing data.

TECHNICAL SKILLS

- **NPI & quality**: APQP, PPAP, PFMEA, DFMEA, control plans, poka-yoke, SPC, Cpk, Gauge R&R (MSA), DMAIC, 5 Why, 8D
- **Data, simulation & CAD**: Power BI, SQL, PostgreSQL, Odoo ERP, SAP, Excel (advanced), Minitab (coursework), JMP (project-level), R, FlexSim, AutoCAD, SolidWorks, Moldflow
- **Lean & planning**: Theory of Constraints, critical path method (CPM), SMED, Kaizen, 5S, value stream mapping, Kanban
- **Certifications**: Six Sigma Green Belt; ISO 9001 Internal Auditor; IATF 16949 Internal Auditor; PMP (in progress)